Q-Learning Based Adaptive Traffic Signal Control Using Lane Area Detectors

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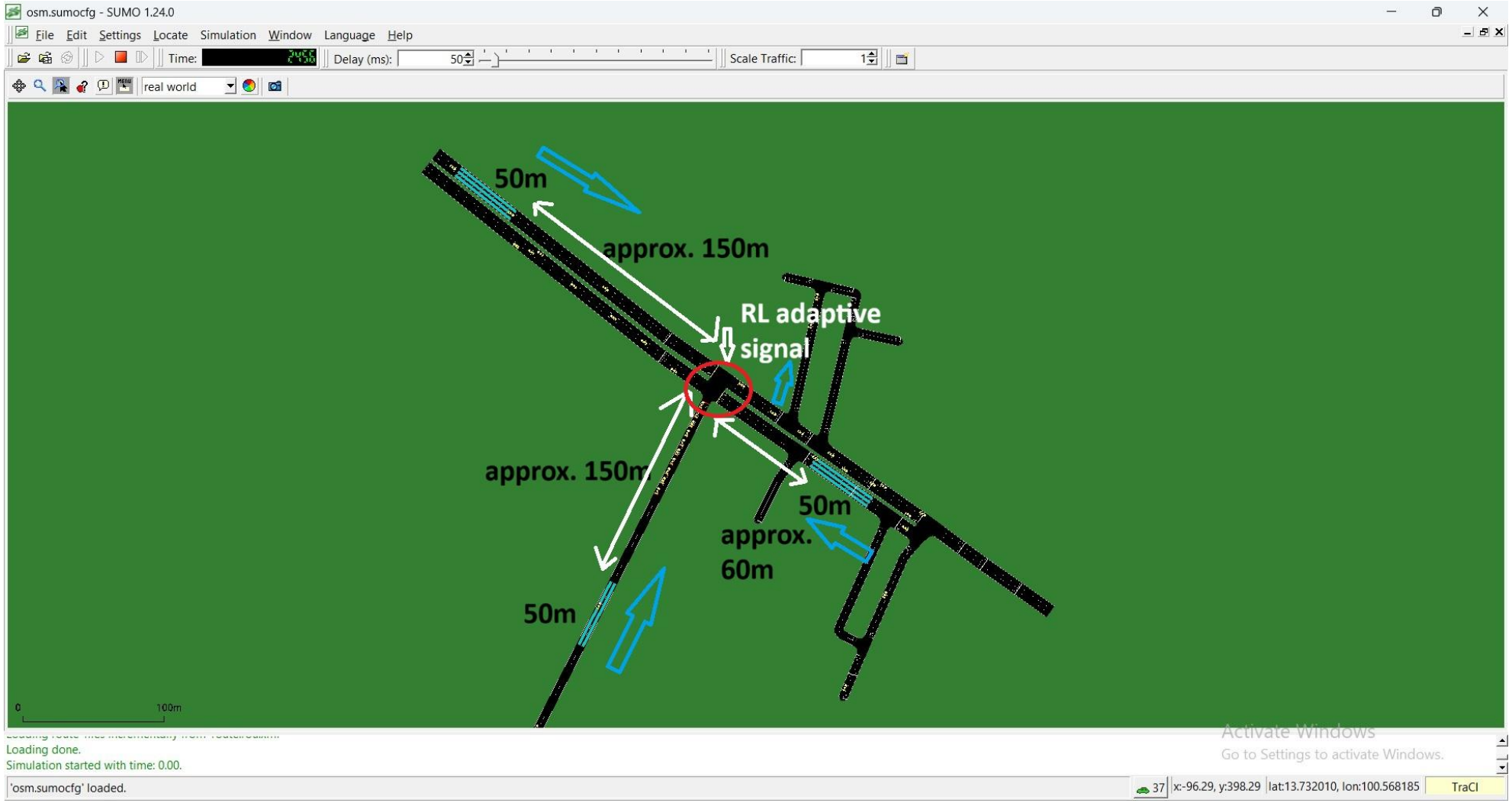
Simulation with .sumocfg



Printing check from TraCI in python console

[STEP 312]
Detectors -> A:1, B:2, C:1
State: (1, 1, 1)
>> Exploiting
Action: 0 (Main Road Green)
Next State: (1, 1, 1)
Reward (−queue): -1
Q-value: 0.000 → -0.100
Max Q(state): 0.000

[STEP 2184]
Detectors -> A:1, B:1, C:1
State: (1, 1, 1)
>> Exploiting
Action: 2 (Side Road Green)
Next State: (1, 1, 1)
Reward (−queue): 0
Q-value: -0.130 → -0.128
Max Q(state): -0.128

TraCI code explanation

Pseudo Code

1. Initialize SUMO, traffic light ID, phases, detectors, RL parameters, Q-table
2. Start simulation
3. For each episode:
4. step ← 0
5. While step < max_steps:
6. Observe detector counts (A, B, C)
7. Discretize counts → state
8. If state not in Q → initialize Q[state]
9. Choose action using ε-greedy (explore/exploit)
10. Apply GREEN phase, then YELLOW phase
11. Advance simulation time
12. Observe next_state and compute reward (−queue length)
13. Update Q(state, action) using Bellman equation
14. state ← next_state, step ← step + phase_duration
15. End simulation

Explanation

1.Initialization

The program sets up the SUMO simulation, traffic light phases, detectors, and Q-learning parameters (learning rate, discount factor, exploration rate). It also creates an empty Q-table to store learned values.

2.State Representation

Traffic conditions are observed using detectors (A, B, C). The number of vehicles is grouped into categories (low, medium, high), forming a discrete state like `(1,2,0)`.

3.Action Selection (ε-greedy)

The agent chooses a signal phase:

1. Sometimes randomly (exploration)
2. Mostly based on the best Q-value (exploitation)

4.Environment Interaction

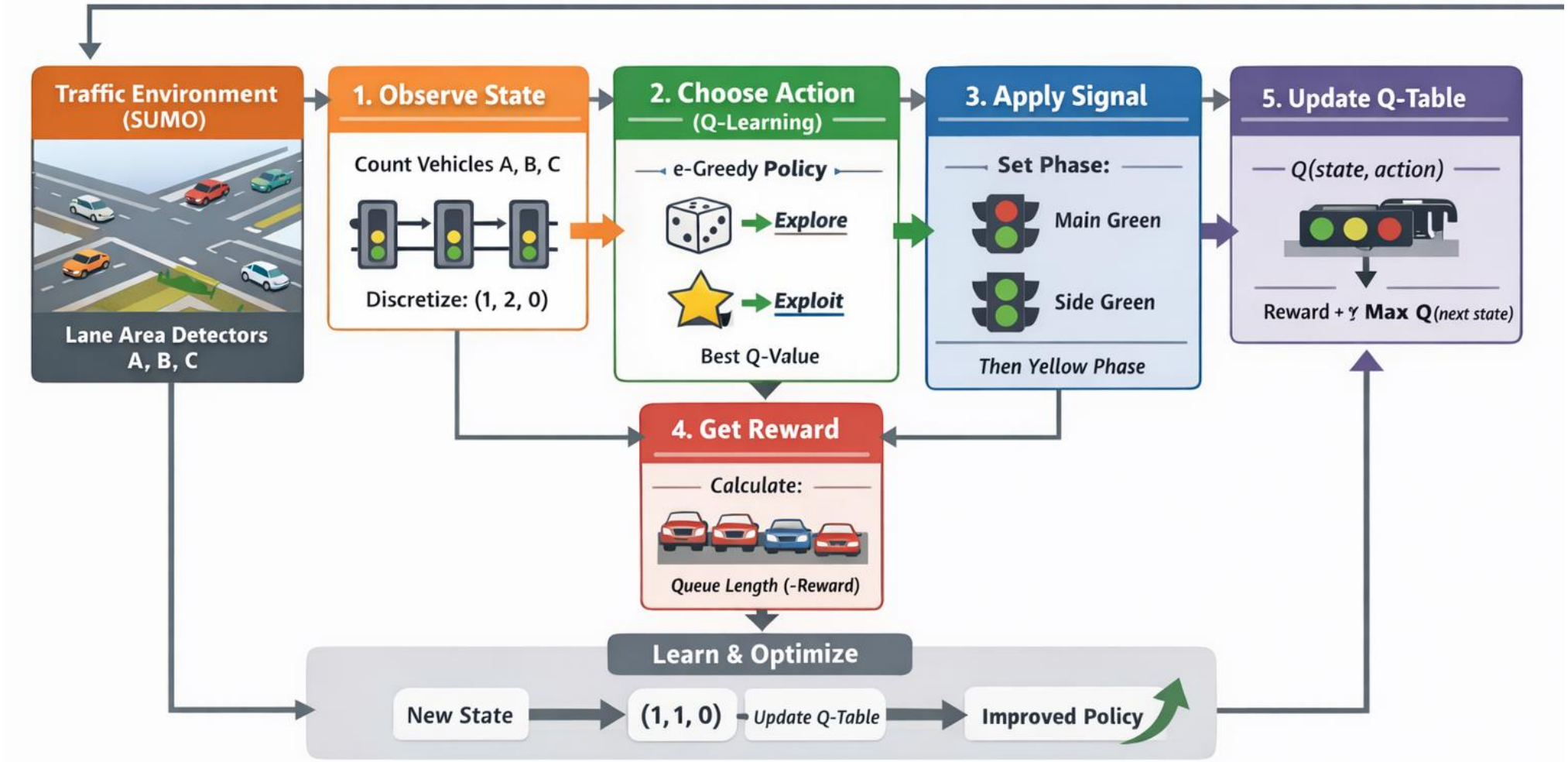
The selected action is applied by setting a green phase followed by a yellow phase. The simulation advances, and new traffic conditions are observed.

5.Learning (Q-update)

The agent receives a reward based on queue length (less queue = better). It updates the Q-table using the Bellman equation so future decisions improve over time.

Reinforcement Learning Adaptive Signal Control

Using Lane Area Detectors



Simulation

GitHub: https://github.com/sauravbarua021-eng/Q-Learning_Adaptive_Traffic_Signal-Control_Lane_area_detectors